EDUTECH STEM
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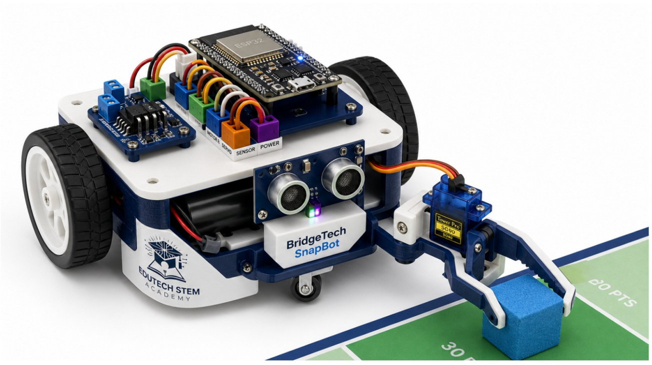
EduTech STEM Academy

Junior Robotics Camp

6-Day Student Lesson Guide and Take-Home Workbook



Name: ______________________________	**Cycle:** __________________
Age: ________	**Team:** __________________

This printable guide is adapted from the EduTech Academy 10-day robotics plan and condensed into a focused 6-day camp for students ages 8 to 16. Students should bring this guide to camp each day and complete the take-home activity after each session.

How To Use This Guide

- Bring this booklet, a pencil, and your best problem-solving attitude to every session.
- Tick each build step only when your team has tested it.
- Use the note spaces to record mistakes, fixes, ideas, and questions.
- Complete the take-home activity after each camp day. It is short, but it helps you think like an engineer.
- Keep the safety rules visible when working with batteries, motors, and moving parts.

Camp Mission

By the end of the camp, each team will build a phone-controlled robot that can move around a tournament arena, grip a small foam cube or ball, carry it, and place it into a scoring zone.

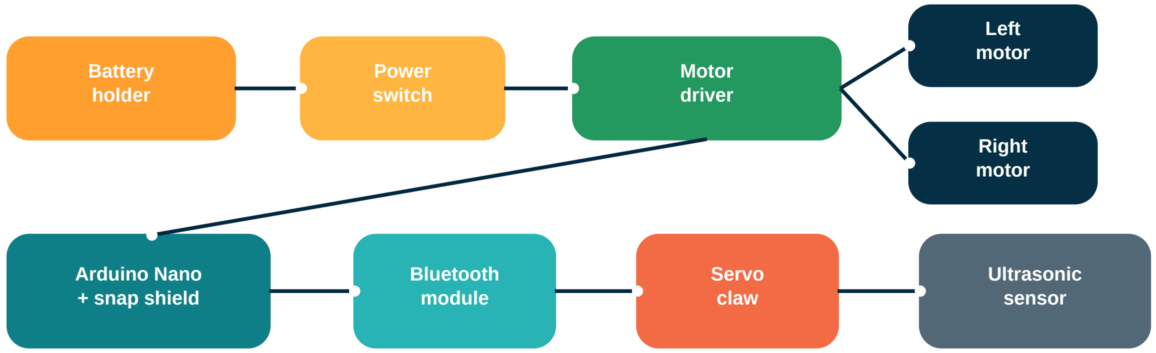
Big idea: information travels from the phone to the Arduino brain. The Arduino tells motors and servos what to do.

Safety Rules

[ ] Do not connect or disconnect batteries unless an instructor gives permission.
[ ] Keep fingers, hair, loose clothing, and jewellery away from spinning wheels and gears.
[ ] If a wire, battery, motor, or board becomes hot, stop immediately and call an instructor.
[ ] Use the power switch before touching moving parts.
[ ] Carry the robot with two hands and place it flat on the table or arena.
[ ] Treat tools, screws, sensors, and teammates with care.

Kit Checklist

[ ] Arduino Nano	[ ] Nano snap-on expansion board
[ ] Motor driver module	[ ] 2 TT gear motors
[ ] 2 wheels	[ ] Front caster wheel
[ ] Servo motor	[ ] 3D printed gripper
[ ] Ultrasonic sensor	[ ] Bluetooth module
[ ] Battery holder	[ ] Power switch
[ ] JST/snap-on cables	[ ] Screws, nuts, spacers



Snap-on rule: match the connector shape, wire color, and port label before pushing gently into place.

Before Powering Up

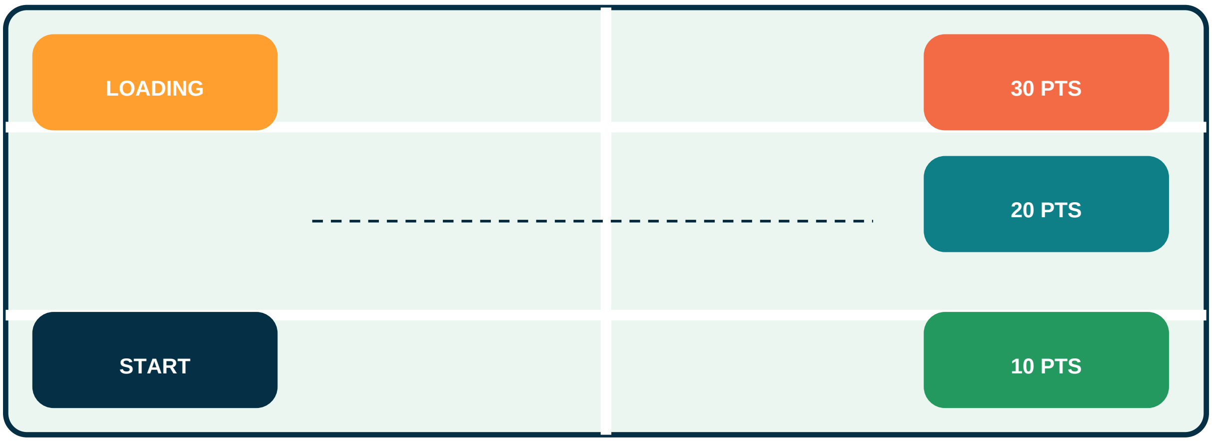
A good robotics team checks the simple things first. Use this space to write the port labels your team will check before turning on the robot.

Ports, wire colours, and questions to ask an instructor

Simple Camp Curriculum Guide

Day	Main Focus	What Students Take Home
1	Scratch mission game and robot logic	A mission algorithm and robot design sketch
2	mBlock, Arduino Nano, and snap-on electronics	A labelled brain-and-wiring diagram
3	Chassis, motors, power, and precision driving	A movement map and troubleshooting log
4	Bluetooth phone control and driver practice	A command table and safe-driver checklist
5	Gripper, sensor ideas, and tournament practice	A strategy plan and gripper improvement idea
6	Final tune-up, mini tournament, and showcase	A take-home care card and extension challenge

Mini tournament arena sketch



Mission: move foam cubes from the Loading Zone into scoring zones. Higher zones need more control.

Team Roles

Role	Responsibility
Builder	Assembles the chassis, checks screws, and keeps parts organized.
Coder	Builds and tests the Scratch or mBlock code with help from teammates.
Driver	Practices smooth phone control and follows the match rules.
Engineer	Records problems, fixes, test results, and design changes.

Day 1 - Scratch Mission Game and Robot Logic

Learn how a robot mission can be planned first as a game, then turned into real movement.

Camp goals	- Understand sprites, stage coordinates, events, loops, and if-then logic. - Create a simple digital tournament mission in Scratch. - Explain how computer-game movement relates to a physical robot.
Materials	- Scratch or Scratch-style workspace - Tournament arena sketch - Pencil - Team planning sheet
Success check	Your Scratch robot sprite moves through a lane, reaches a scoring zone, and stops without leaving the mission area.

Daily lesson flow

Time	Activity
9:00	Welcome, safety expectations, team roles, and preview of the final mini tournament.
9:30	Scratch mini lesson: sprites, coordinates, keyboard events, loops, and boundaries.
10:15	Team build: create a robot sprite mission game with a start zone, lane, loading zone, and scoring zone.
12:00	Lunch/break and quick team share.
12:30	Improve the game with rules: penalty zone, score counter, timer, or obstacle.
1:30	Test block: demonstrate the mission and explain one logic rule.
1:50	Reflection and take-home task.

Build and code steps

[ ] Open Scratch and create or choose a robot sprite.
[ ] Draw or import a simple arena background with a Start Zone and Scoring Zone.
[ ] Add arrow-key movement using event blocks.
[ ] Add boundary rules so the robot cannot leave the arena.
[ ] Add a score rule when the robot reaches the scoring zone.
[ ] Test the game three times and fix one bug.
[ ] Write one sentence explaining how your Scratch command could become a real robot command.

Day 1 - Scratch Mission Game and Robot Logic - Take-Home Workbook Page

Use this page after camp to slow down, think, and prepare for the next robotics session.

Student notes and take-home assignment

Reflection question	What was easier to control in Scratch than it will be with a real robot? Why?
Homework	At home, find one machine that follows instructions. Draw it and write three commands it might follow, such as start, stop, turn, sense, or lift.
Parent/guardian sign-off	Name: ______________________________ Signature: ______________________________

My notes, sketch, or questions for tomorrow

Day 2 - mBlock, Arduino Nano, and Snap-On Electronics

Meet the robot brain and learn how block code can control real hardware.

Camp goals	- Identify the Arduino Nano, snap-on expansion board, ports, and safe cable direction. - Connect the Nano to mBlock and upload a simple test program. - Use LED or built-in blink tests to prove that code is reaching the robot brain.
Materials	- Arduino Nano - Nano expansion board - USB cable - mBlock - LED module or built-in LED
Success check	Your team changes the blink timing and uploads the code successfully to the Arduino Nano.

Daily lesson flow

Time	Activity
9:00	Review homework and connect Scratch logic to robot logic.
9:25	Parts tour: Arduino Nano, expansion board, ports, power, sensors, and outputs.
10:00	mBlock mini lesson: connect board, choose device, upload mode, and run a blink test.
11:15	Team test: change blink speed and record what happened.
12:00	Lunch/break.
12:30	Snap-on electronics practice: connect and remove cables correctly with instructor checks.
1:25	Robot coding license test: each team uploads a changed program.
1:50	Reflection and take-home task.

Build and code steps

[ ] Place the Arduino Nano on the snap-on board exactly as shown by the instructor.
[ ] Connect the USB cable to the computer before connecting battery power.
[ ] Open mBlock and select the correct Arduino Nano device or compatible board option.
[ ] Upload the starter blink program.
[ ] Change the delay time and upload again.
[ ] Label one input, one output, and one power connection on your worksheet.
[ ] Ask for an instructor check before using any battery pack.

Day 2 - mBlock, Arduino Nano, and Snap-On Electronics - Take-Home Workbook Page

Use this page after camp to slow down, think, and prepare for the next robotics session.

Student notes and take-home assignment

Reflection question	What steps helped your team solve connection problems?
Homework	Label the robot system diagram in this booklet. Circle the brain, underline one input, box one output, and write one safety rule beside the battery.
Parent/guardian sign-off	Name: ______________________________ Signature: ______________________________

My notes, sketch, or questions for tomorrow

Day 3 - Chassis, Motors, Power, and Precision Driving

Build the drivetrain and make the robot move with control, not just speed.

Camp goals	- Assemble chassis, wheels, motors, caster, battery holder, and motor driver. - Understand how the motor driver controls direction and speed. - Drive forward, backward, left, and right inside a test lane.
Materials	- Chassis - 2 TT motors - 2 wheels - Caster wheel - Motor driver - Battery pack - Snap cables
Success check	Your robot travels 50 cm forward and backward, then completes a controlled left and right turn.

Daily lesson flow

Time	Activity
9:00	Safety check and drivetrain demo: raw battery power versus controlled motor driver power.
9:30	Mechanical build: mount motors, wheels, caster, battery holder, and driver module.
10:45	Wiring check: match motor outputs and snap-on control wires with instructor approval.
11:20	mBlock movement code: forward, backward, stop, left turn, right turn.
12:00	Lunch/break.
12:30	Calibration: adjust speed and delay for straight driving and 90-degree turns.
1:35	Drive test: 50 cm movement and checkpoint turn drill.
1:50	Reflection and take-home task.

Build and code steps

[ ] Mount both motors firmly and make sure the wheels spin freely.
[ ] Install the caster wheel at the front or rear as directed.
[ ] Mount the battery holder and motor driver where wires will not touch wheels.
[ ] Connect motor wires to the motor driver with instructor help.
[ ] Use snap-on signal cables between the motor driver and expansion board.
[ ] Write or load movement blocks for forward, backward, stop, left, and right.
[ ] Test at low speed first, then tune speed and timing.

[ ] Record one problem and the fix in your troubleshooting log.

Day 3 - Chassis, Motors, Power, and Precision Driving - Take-Home Workbook Page

Use this page after camp to slow down, think, and prepare for the next robotics session.

Student notes and take-home assignment

Reflection question	How did friction, battery power, or wheel direction affect your robot?
Homework	Draw a simple movement map from Start to Score. Mark where the robot must go forward, turn, slow down, grip, reverse, or stop.
Parent/guardian sign-off	Name: ______________________________ Signature: ______________________________

My notes, sketch, or questions for tomorrow

Day 4 - Bluetooth Phone Control and Driver Practice

Turn the robot into a wireless machine and practise safe, smooth driving.

Camp goals	- Pair a phone with the robot using Bluetooth. - Connect single-character phone commands to robot movement blocks. - Complete a safe driver license test using forward, backward, left, right, and stop.
Materials	- Bluetooth module - Android phone control app - Arduino Nano robot - Test lane - Driver checklist
Success check	Your robot responds to phone commands quickly and stops safely when the stop command is pressed.

Daily lesson flow

Time	Activity
9:00	Review movement maps and choose team driver/coder/builder roles.
9:25	Bluetooth mini lesson: pairing, serial commands, and the F, B, L, R, S command pattern.
10:00	Connect Bluetooth module with instructor check.
10:35	mBlock logic tree: if command equals F, move forward; if command equals S, stop.
12:00	Lunch/break.
12:30	Phone pairing and controlled driving drills.
1:25	Driver license test: complete movement commands and immediate stop.
1:50	Reflection and take-home task.

Build and code steps

[ ] Power off the robot before plugging in the Bluetooth module.
[ ] Match the Bluetooth cable to the correct snap-on port as shown by the instructor.
[ ] Pair the phone with the correct module name.
[ ] Build the command table: F forward, B backward, L left, R right, S stop.
[ ] Test one command at a time at low speed.
[ ] Add or confirm a stop command before full driving.
[ ] Complete a no-crash lap inside the test lane.

Day 4 - Bluetooth Phone Control and Driver Practice - Take-Home Workbook Page

Use this page after camp to slow down, think, and prepare for the next robotics session.

Student notes and take-home assignment

Reflection question	What makes a good robot driver: speed, smoothness, planning, or teamwork? Explain your answer.
Homework	Write your phone command table. Add one extra command you would like to create, such as open claw, close claw, slow mode, or celebrate LED.
Parent/guardian sign-off	Name: ____________________________ Signature: ____________________________

My notes, sketch, or questions for tomorrow

Day 5 - Gripper, Sensor Ideas, and Tournament Practice

Add the robot hand, improve reliability, and practise the mission under pressure.

Camp goals	- Mount and test a servo-powered gripper or scoop. - Use open and close commands to capture a foam cube or ball. - Explore sensors and LED feedback as optional robot upgrades.
Materials	- SG90 servo - 3D printed gripper - Foam cube or ball - Ultrasonic sensor - LED module - Tournament arena
Success check	Your robot can approach one object, grip or trap it, reverse, and place it into a scoring zone.

Daily lesson flow

Time	Activity
9:00	Review phone command homework and choose today's improvement goal.
9:25	Servo mini lesson: angles, open position, closed position, and gentle mechanical limits.
10:00	Mount gripper and connect servo to the correct snap-on servo port.
10:45	Code open and close commands, then test without touching the object first.
12:00	Lunch/break.
12:30	Object capture practice and optional sensor/LED status upgrade.
1:25	Practice match: two-minute mission with score and penalty tracking.
1:50	Reflection and take-home task.

Build and code steps

[ ] Inspect the gripper parts and identify where the servo horn attaches.
[ ] Mount the gripper without blocking the wheels or caster.
[ ] Plug the servo cable into the correct signal, power, and ground order.
[ ] Set safe open and closed angles in mBlock.
[ ] Test open and close while the robot is lifted off the arena.
[ ] Practise approaching the object slowly before closing the claw.
[ ] Run one timed practice match and record the score.

[ ] Choose one improvement: grip padding, slower approach, better wire routing, or driver strategy.

Day 5 - Gripper, Sensor Ideas, and Tournament Practice - Take-Home Workbook Page

Use this page after camp to slow down, think, and prepare for the next robotics session.

Student notes and take-home assignment

Reflection question	What failed during the first full mission test, and what did your team change?
Homework	Design one gripper improvement or tournament strategy. Draw it, name it, and explain how it could earn more points or reduce penalties.
Parent/guardian sign-off	Name: ______________________________ Signature: ______________________________

My notes, sketch, or questions for tomorrow

Day 6 - Final Tune-Up, Mini Tournament, and Showcase

Use everything you learned to compete, explain your design, and plan your next robotics project.

Camp goals	- Complete final inspection for wiring, battery, stop command, and structure. - Compete in qualification and final matches with teamwork and sportsmanship. - Explain one test, one failure, and one improvement made by the team.
Materials	- Completed robot - Tournament arena - Score sheet - Inspection checklist - Certificate or award sheet
Success check	Your team competes safely, records a score, explains the design, and leaves with a take-home challenge.

Daily lesson flow

Time	Activity
9:00	Final hardware tight-check, battery check, code backup, and driver warm-up.
9:40	Pre-match inspection: size, wires, safe power switch, stop command, and gripper movement.
10:15	Engineering interview: explain design choices, coding logic, tests, failures, and fixes.
11:00	Qualification rounds: timed scoring runs.
12:00	Lunch/break and bracket update.
12:30	Finals round and sportsmanship awards.
1:30	Showcase, reflection, certificates, and take-home care briefing.
1:55	Pack robot safely and confirm parent/guardian collection.

Build and code steps

[ ] Tighten loose screws and make sure no wire can touch wheels or gears.
[ ] Run movement, stop, open, and close commands before the match.
[ ] Complete the inspection checklist with an instructor.
[ ] Prepare two speakers for the engineering interview.
[ ] Run qualification matches and record scores honestly.
[ ] Celebrate good driving, teamwork, and improvement, not only winning.
[ ] Write your next robotics goal before leaving camp.

Day 6 - Final Tune-Up, Mini Tournament, and Showcase - Take-Home Workbook Page

Use this page after camp to slow down, think, and prepare for the next robotics session.

Student notes and take-home assignment

Reflection question	What are you most proud of building, coding, fixing, or learning during camp?
Homework	Complete the 7-day take-home extension challenge. Each day, improve one small part of your robot plan, code idea, driving skill, or engineering notebook.
Parent/guardian sign-off	Name: ______________________________ Signature: ______________________________

My notes, sketch, or questions for tomorrow

Worksheets and Score Pages

Robot Design Sketch

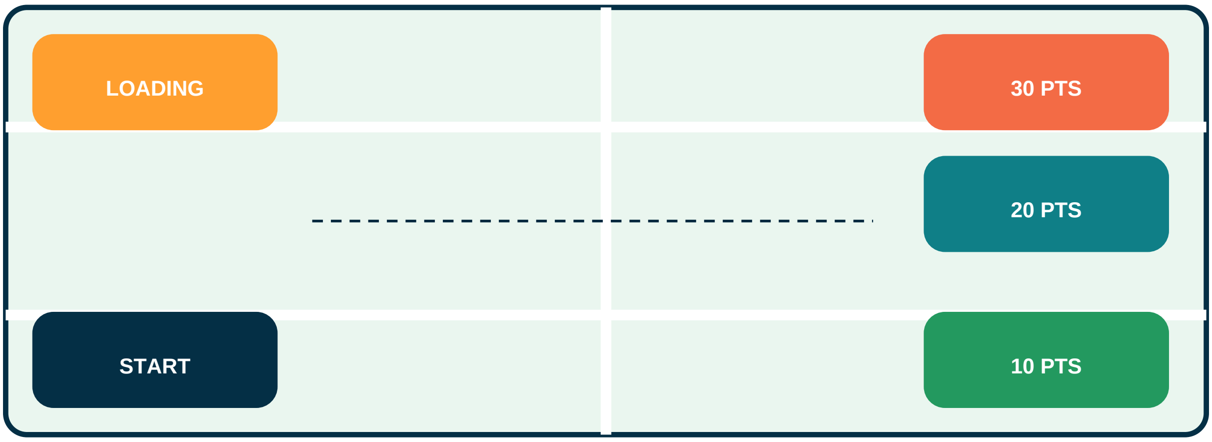
Draw your robot from the top view. Label the brain, battery, wheels, gripper, sensor, and power switch.

Troubleshooting Log

Problem	Possible Cause	Fix Tested	Result

Mini Tournament Score Sheet

Mini tournament arena sketch



Mission: move foam cubes from the Loading Zone into scoring zones. Higher zones need more control.

Scoring Item	Value	Count	Total
Object delivered to 10-point zone	**10 pts**		
Object delivered to 20-point zone	**20 pts**		
Object delivered to 30-point zone	**30 pts**		
Completed match with no human touch	**+10 bonus**		
Team explained one test and one improvement	**+10 bonus**		
Human touch during match	**-5 penalty**		
Robot leaves arena boundary	**-10 penalty**		
Unsafe driving or ignoring stop instruction	**-15 penalty**		
Final Score			

Engineering Interview Practice

Write one test, one failure, one fix, and one thing your team would improve next.

7-Day Take-Home Extension Challenge

[ ] Day 1: Draw a cleaner wire layout.

[ ] Day 2: Write a safer driving checklist.

[ ] Day 3: Improve your command table with one new command.
[ ] Day 4: Redesign the gripper for better control.
[ ] Day 5: Sketch a sensor-based autonomous idea.
[ ] Day 6: Teach someone at home what an input and output are.
[ ] Day 7: Write the next robot you want to build.

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